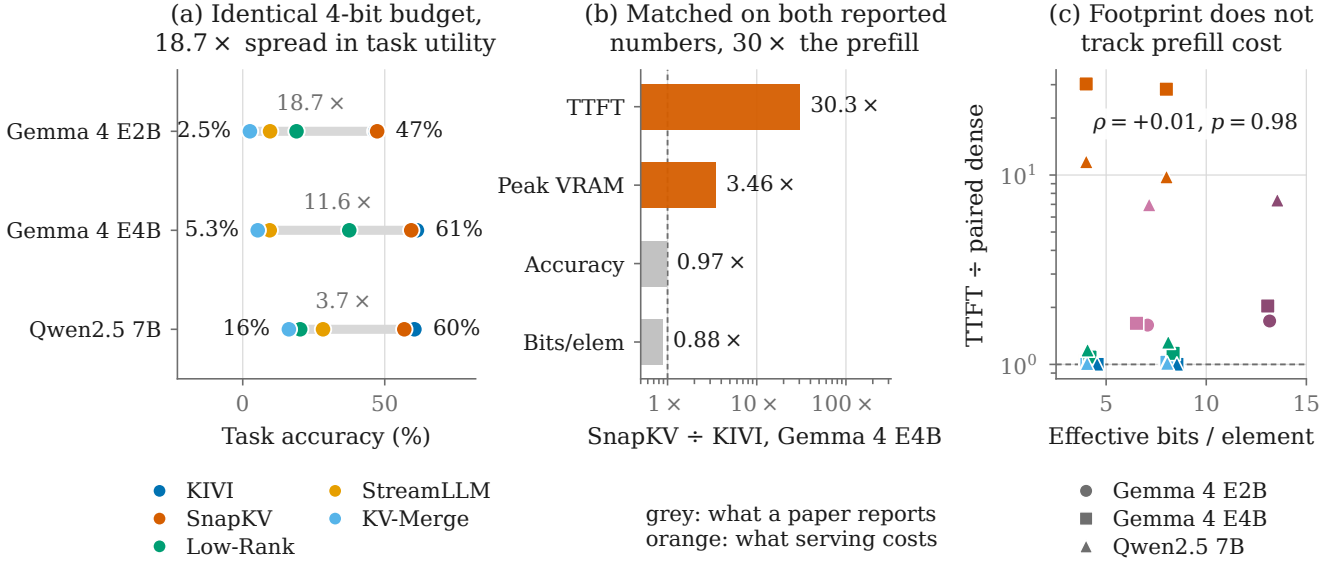
(a) Identical 4-bit budget,
18.7 × spread in task utility
Gemma 4 E2B
Gemma 4 E4B
Qwen2.5 7B
18.7 ×
11.6 ×
3.7 ×
2.5%
47%
5.3%
61%
16%
60%
0
50
Task accuracy (%)
KIVI
SnapKV
Low-Rank
StreamLLM
KV-Merge
(b) Matched on both reported
numbers, 30 × the prefill
TTFT
Peak VRAM
Accuracy
Bits/elem
30.3 ×
3.46 ×
0.97 ×
0.88 ×
1 ×
10 ×
100 ×
SnapKV ÷ KIVI, Gemma 4 E4B
grey: what a paper reports
orange: what serving costs
(c) Footprint does not
track prefill cost
ρ = +0.01, p = 0.98
TTFT ÷ paired dense
10^1
10^0
5
10
15
Effective bits / element
Gemma 4 E2B
Gemma 4 E4B
Qwen2.5 7B